Performance of State-of-the-Art Multimodal Large Language Models on an Image-Rich Radiology Board Examination: Comparison to Human Examinees

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Rationale and Objectives: This study aimed to assess the current multimodal capabilities of leading multimodal large language models (MLLMs) using a 2024 radiology board examination, evaluate their proficiency in utilizing medical image content, compare their performance against human examinees, and consider their cost-effectiveness.

Material and Methods: Six contemporary MLLMs (GPT-4.1, o3, Claude 3.7 Sonnet, Claude 3.7 Sonnet-thinking, Gemini 2.5 Pro Preview, and Gemini 2.5 Flash Preview-thinking) were evaluated using the 100 multiple-choice questions (96 image-based) from the 2024 official board examination of the Japan Radiological Society. Questions, originally in Japanese, were instructed to be translated into English by the MLLMs. Performance was also analyzed with and without images for certain models to assess multimodal utility.

Results: Gemini 2.5 Pro Preview achieved the highest accuracy (76.0%), followed by o3 (75.0%), both surpassing the average human examinee score (72.9%). Gemini 2.5 Pro Preview showed 75.0% accuracy with images versus 63.5% without ($p = 0.035$), and Gemini 2.5 Flash Preview-thinking demonstrated 68.8% accuracy with images versus 57.3% without ($p = 0.019$), indicating significant performance gains with image inclusion. Notably, Gemini models demonstrated top-tier performance at a highly competitive cost.

Conclusion: The latest generation of MLLMs, particularly Gemini 2.5 Pro Preview and o3, can exceed average human performance on radiology board examinations and effectively leverage image information. The Gemini series, in particular, shows rapid improvements and offers a compelling combination of high performance and cost-efficiency for potential applications in radiology.

Summary Statement: Modern multimodal large language models, notably Gemini 2.5 Pro Preview and o3, surpassed average human performance on the 2024 Japanese Radiology Board Examination. Gemini models showed significant score improvements when utilizing image data and offer top-tier performance at a competitive cost, indicating rapid advancements and excellent cost-effectiveness for radiology applications.

Key Words: Large language models (LLMs); Multimodal AI; Radiology Board Examination; Medical image analysis; Cost-effectiveness.

Abbreviations: AI Artificial Intelligence, **API** Application Programming Interface, **CT** Computed Tomography, **JRS** Japan Radiological Society, **LLM** Large Language Model, **MLLM** Multimodal Large Language Model, **MRI** Magnetic Resonance Imaging, **MSK** Musculoskeletal, **PACS** Picture Archiving and Communication Systems, **PDF** Portable Document Format, **RI** Radioisotope, **RLHF** Reinforcement Learning from Human Feedback, **SD** Standard Deviation

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INTRODUCTION

The field of medicine has witnessed a transformative impact from the rapid advancements in deep learning, particularly through the development of large language models (LLMs). These sophisticated artificial intelligence (AI) systems have demonstrated remarkable capabilities in processing and understanding complex textual information, leading to notable successes where LLMs have achieved high, sometimes passing or near-passing, scores on various medical licensing and specialty board examinations (1,2). Building on this, multimodal LLMs (MLLMs), which can concurrently process diverse data types such as text and images, have emerged with significant promise for image-intensive medical fields like radiology. In radiology, diagnostic interpretation intrinsically relies on a nuanced understanding of visual data within a clinical context. However, the initial application of MLLMs in this domain encountered challenges. Early studies often indicated that the inclusion of radiological images with textual questions did not consistently lead to performance improvements in tasks like specialty board examinations, suggesting that these earlier MLLMs struggled to effectively interpret complex medical visual data despite their textual proficiency (3–5). The potential of AI in radiologic reporting and examination systems has been previously reported (6,7). However, to advance such clinical applications, rigorous validation of model accuracy in radiology—through benchmarks such as board examinations—remains a critical foundation.

The development of LLMs and MLLMs is, however, characterized by continuous and rapid evolution. Recent models have improved their reasoning abilities, advancing from pattern recognition to multi-step problem-solving and logical deduction. These gains have been driven by refinements in model design and training methods, such as instruction tuning and reinforcement learning (8,9). At the same time, multimodal learning has advanced, with newer models designed to integrate image and text data from the start. Innovations in vision encoders and training on diverse multimodal datasets now enable a more coherent understanding of images and text (10,11). These ongoing efforts suggest that newer MLLMs might now be better equipped to overcome the previous limitations in handling radiological image-based problems. Given this landscape of rapid technological progress and the previously identified challenges, an up-to-date evaluation of the newest generation of MLLMs in the context of radiology is crucial.

The purpose of this study is to evaluate the performance of leading contemporary MLLMs (defined as models released in early 2025 and representing flagship offerings from OpenAI, Anthropic, and Google) on the 2024 Japanese Radiology Board Examination. Unlike prior work, this study directly compares the newest generation of models against average human specialist performance on an image-rich national certification exam, while also assessing the added value of multimodal input and cost-effectiveness. By providing a contemporary benchmark, we aim to highlight the extent to which MLLMs have matured in radiology and their potential impact on education and clinical practice.

MATERIALS AND METHODS

Ethical Considerations

This study used publicly available exam questions and did not involve human or animal subjects; therefore, ethical approval was not required. The data used were anonymized and handled in accordance with relevant privacy regulations and guidelines.

Data Collection

The dataset comprised 100 multiple-choice questions from the 2024 official board examination of the Japan Radiological Society (JRS), 2nd stage. The JRS 2nd stage exam is a high-stakes national certification for radiology residents in Japan, typically taken in postgraduate year 5. Its structure, difficulty, and breadth are comparable to major international exams such as the ABR Core (US) and FRCR (UK), focusing heavily on multimodal image interpretation. Of the 100 questions, 96 (96%) included radiological images, while four (4%) were text-only. The questions, originally in Japanese, covered six subspecialties as detailed in Table 1.

The examination questions were downloaded in Portable Document Format (PDF) from a members-only section of the JRS website. Text and image information were extracted from each question's PDF file using Adobe Acrobat (Adobe, San Jose, CA, USA). The textual portions of the questions were copied and pasted into simple text files. The image portions, limited to the key images provided for each question, were separated into individual PNG images at 300 DPI.

Performance data for human radiologists (fifth-year residents) were obtained through inquiry to the JRS, revealing an average examination accuracy of 72.9% (Standard Deviation [SD] = 9.26%).

Model Selection

Six contemporary MLLMs, representing the state-of-the-art as of Spring 2025 (based on their release dates detailed in Table 2), were selected for this evaluation. The selection was guided by the following two principles: 1) including the latest flagship models from three prominent commercial AI developers (OpenAI, Anthropic, and Google), and 2) ensuring diversity across architectures and available variants. In this context, we did not impose a strict categorization of models as reasoning versus non-reasoning, since recent releases often blur such distinctions. Instead, we aimed to capture the breadth of currently available approaches, including developer-labeled 'thinking' variants where offered. The chosen models were OpenAI's GPT-4.1 and o3; Anthropic's Claude 3.7 Sonnet and Claude 3.7 Sonnet-

TABLE 1. Distribution of Questions by Subspecialty in the Japanese Radiology Board Examination (Second stage)

Subspecialty	Number of Questions	Image-based Questions (%)
Abdominal and Pelvic Radiology	27	27 (100%)
Thoracic and Cardiac Radiology	20	20 (100%)
Nuclear Medicine	20	20 (100%)
Neuroradiology and Head and Neck Imaging	16	16 (100%)
Musculoskeletal and Breast Imaging	11	11 (100%)
Other	6	2 (33.3%)
Total	*100*	*96 (96%)*

Notes: Data are from the second-stage examination of the Japanese Radiology Board Certification for fifth-year radiology residents. The exam comprised 100 multiple-choice questions, 96 of which were image-based.

thinking; and Google's Gemini 2.5 Pro Preview and Gemini 2.5 Flash Preview-thinking. These models are referenced in this paper by these common names, and their OpenRouter Application Program.

MLLM Interrogation and Data Processing

Interactions with the MLLMs via the OpenRouter API (https://openrouter.ai/) were scripted using Python (version 3.11.7, https://www.python.org/) and the openai library (version 1.63.2) to automate the testing and evaluation process. Each model was queried once per question without retries to simulate a standardized examination setting. All API calls were stateless, with no conversation memory retained across queries. Each question was processed independently to prevent potential carry-over of information or learning effects between items.

For the 96 image-based questions, two conditions were tested as follows:

With Image (Multimodal Input):

- All associated radiological images were extracted as PNG format without cropping, maintaining the original resolution. Images were converted to Base64-encoded strings using Python's base64.b64encode() function to meet API requirements. No preprocessing steps, such as normalization, contrast adjustment, or enhancement were applied.
- The MLLM was prompted with: "As a radiologist, you are required to provide the answer to this multiple-choice question. Also, since the question is written in Japanese, please translate it into English before solving the question. Image files are also attached separately. No explanation of the content is necessary; simply answer using “a”, “b”, “c”, “d”, and “e.”
- The text prompt and the Base64-encoded image were sent in the API request. For this multimodal input, the request to the MLLM included both the text prompt and the image data. The image was formatted as a data URI (specifically, data: image/png; base64, followed by the Base64-encoded image string) within the designated field for image URLs in the API's message structure, allowing the model to process both text and image simultaneously.

Without Image (Text-only Input):

- The same 96 image-based questions were presented to the MLLMs again, but without the image data.
- The prompt used was: "As a radiologist, you are required to provide the answer to this multiple-choice question. Also, since the question is written in Japanese, please translate it into English before solving the question. No explanation of the content is necessary; simply answer using “a”, “b”, “c”, “d”, and “e.”

The four text-only questions were processed using the "Without Image" prompt and procedure. The overall study workflow, including this procedure, is illustrated in Figure 1.

This prompt format was chosen to replicate a board examination setting, where each question is answered independently and only a single choice is required. Using a uniform and minimal instruction reduced variability across models and ensured fairness.

Responses from the MLLMs were received via the API. In the majority of cases, model outputs included not only a single-letter answer but also explanatory text. To ensure consistency and avoid bias, all such responses were manually reviewed and converted into the corresponding single-letter choice before accuracy evaluation. This ensured that all responses were standardized for accurate evaluation. If a model explicitly declined to answer a question (e.g., indicating that an image was required), the response was classified as incorrect under the strict scoring protocol.

Statistical Analysis

The primary outcome measure was accuracy, defined as the percentage of correctly answered questions. Scoring was based on a strict exact match criterion; only responses perfectly matching the correct answer key were considered correct, and no partial credit was given. Overall accuracies and subspecialty accuracies were calculated for each MLLM and compared to the average performance of human examinees.

McNemar's test was applied exclusively to paired results of the 96 image-based questions, comparing with-image versus without-image performance. McNemar's test was selected as

TABLE 2. Specifications and Pricing of Representative Large Language Models

Model Name	Vendor	OpenRouter Identifier	Release Date	Knowledge Cut-off	Cost ($/1M tokens)
GPT-4.1	OpenAI	openai/gpt-4.1	2025-04-14	Oct 2023	2.00 / 8.00
o3	OpenAI	openai/o3	2024-03-15	Oct 2023	10.00 / 40.00
Claude 3.7 Sonnet	Anthropic	anthropic/claude-3.7-sonnet	2025-02-24	Apr 2024	3.00 / 15.00
Claude 3.7 Sonnet-thinking	Anthropic	anthropic/claude-3.7-sonnet:thinking	2025-02-24	Apr 2024	3.00 / 15.00
Gemini 2.5 Pro Preview	Google	google/gemini-2.5-pro-preview	2025-04-09	Jan 2025	1.25 / 10.00
Gemini 2.5 Flash Preview-thinking	Google	google/gemini-2.5-flash-preview:thinking	2025-04-17	Jan 2025	0.15 / 3.50

Notes: Release date and knowledge cut-off details provided in the main text. Pricing information updated based on OpenRouter (https://openrouter.ai/) data as of May 20, 2025.

it is specifically designed for paired binary outcomes, making it well-suited for comparing performance with versus without images for the same set of questions. A p-value less than 0.05 was considered statistically significant. All statistical analyses were performed using Python version 3.11.7 with SciPy, version 1.13.1.

RESULTS

The 2024 official board examination of the JRS (2nd stage) comprised 100 questions, with 96 (96%) being image-based, covering six subspecialties as detailed in Table 1. The average performance of human examinees (fifth-year radiology residents) was 72.9% (SD = 9.26%).

Overall Model Performance

Among the evaluated MLLMs, Gemini 2.5 Pro Preview achieved the highest total accuracy at 76.0%, followed closely by o3 with 75.0%. Both models surpassed the average performance of human examinees (Table 3 and Fig 2). The other models evaluated, Gemini 2.5 Flash Preview-thinking, GPT-4.1, Claude 3.7 Sonnet-thinking, and Claude 3.7 Sonnet, achieved total accuracies of 70.0%, 62.0%, 62.0%, and 61.0%, respectively (Table 3 and Fig 2).

Performance by Subspecialty

As detailed in Table 3 and Figure 3, Gemini 2.5 Pro Preview demonstrated strong performance across various subspecialties, ranking first in Thoracic & Cardiac Radiology (85.0%), Nuclear Medicine (80.0%), and "Other" questions (100.0%). It ranked second in Abdominal & Pelvic Radiology (77.8%). The o3 model achieved the highest accuracy in Abdominal & Pelvic Radiology (81.5%) and ranked second in Neuroradiology & Head/Neck (62.5%), Thoracic & Cardiac Radiology (80.0%), Nuclear Medicine (75.0%), and "Other" (83.3%). Gemini 2.5 Flash Preview-thinking was the top performer in Musculoskeletal & Breast Imaging (MSK & Breast) (81.8%) and shared the top rank in Thoracic & Cardiac Radiology (85.0%) and "Other" (100.0%). GPT-4.1 showed its best relative performance in Neuroradiology & Head/Neck, ranking first with 68.8%.

Impact of Image Data on Performance

The utility of multimodal input was assessed by comparing performance on image-based questions with and without the accompanying images (Table 4 and Fig 4). Gemini 2.5 Pro Preview showed a statistically significant improvement in accuracy when images were provided (75.0% with images vs. 63.5% without images; $p = 0.035$). Similarly, Gemini 2.5 Flash Preview-thinking demonstrated enhanced accuracy with images (68.8% vs. 57.3%; $p = 0.019$). Claude 3.7 Sonnet also significantly benefited from image inclusion (60.4% vs. 41.7%; $p < 0.001$). Notably, for Claude 3.7 Sonnet, in 40 out of 100 questions, and for Claude 3.7

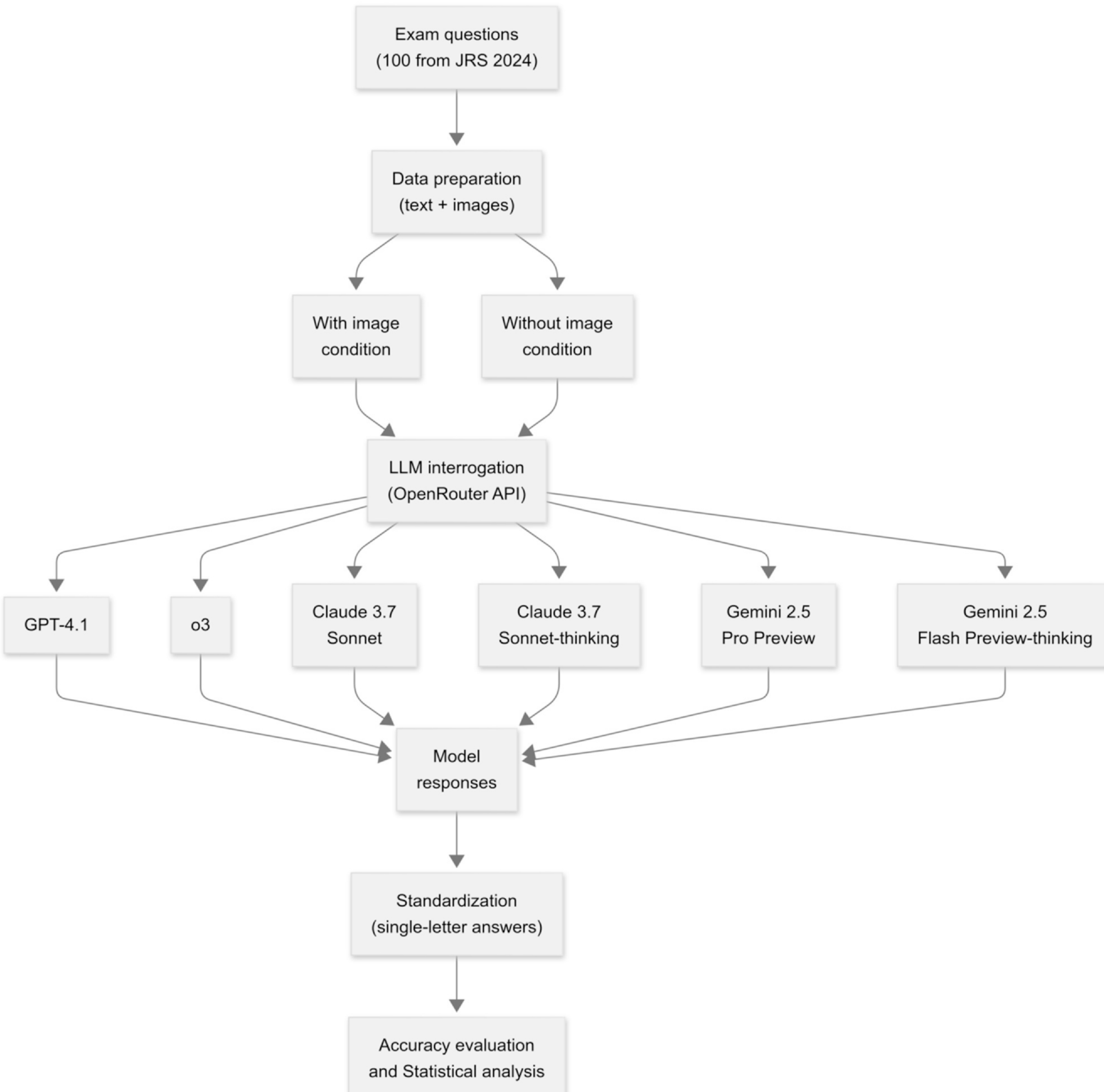

Figure 1. This flowchart illustrates the study design and data analysis process. A total of 100 multiple-choice questions from the 2024 Japanese Radiology Board Examination (JRS, 2nd stage) were extracted. Text and image data were prepared separately and analyzed under two conditions (with image input vs. without image input). Six multimodal large language models (MLLMs)—GPT-4.1, o3, Claude 3.7 Sonnet, Claude 3.7 Sonnet-thinking, Gemini 2.5 Pro Preview, and Gemini 2.5 Flash Preview-thinking—were interrogated via the OpenRouter Application Programming Interface (API). Model responses were collected, standardized to single-letter answers, and compared against the official answer key. Final outputs included overall and subspecialty accuracies as well as statistical testing to assess the impact of image inclusion. API, Application Programming Interface; JRS, Japan Radiological Society; LLM, Large Language Model; MLLM, Multimodal Large Language Model.

Sonnet-thinking, in 11 out of 100 questions, the models responded that they could not provide an answer without images. In contrast, GPT-4.1 (61.5% vs. 63.5%; $p = 0.824$), o3 (74.0% vs. 70.8%; $p = 0.607$), and Claude 3.7 Sonnet-thinking (60.4% vs. 61.5%; $p = 1.000$) did not show statistically significant performance changes with the inclusion of images.

Model Specifications and Cost Considerations

Specifications, including vendor, OpenRouter API identifiers, release dates, knowledge cut-offs, and pricing for the evaluated MLLMs are presented in Table 2 and Figure 5. The costs per 1 million tokens (input/output) varied among models. For instance, Gemini 2.5 Flash Preview-thinking was listed at \$0.15 / \$3.50, while Gemini 2.5 Pro Preview

TABLE 3. Accuracy by Model and Subspecialty in the Japanese Radiology Board Examination

Model Name	Total	Neuroradiology & Head/Neck	MSK & Breast	Thoracic & Cardiac	Abdominal & Pelvic	Nuclear Medicine	Other
Gemini 2.5 Pro Preview	76.0% (1)	56.2% (3)	63.6% (4)	85.0% (1)	77.8% (2)	80.0% (1)	100.0% (1)
o3	75.0% (2)	62.5% (2)	63.6% (4)	80.0% (2)	81.5% (1)	75.0% (2)	83.3% (2)
Gemini 2.5 Flash Preview-thinking	70.0% (3)	50.0% (4)	81.8% (1)	85.0% (1)	63.0% (3)	65.0% (3)	100.0% (1)
GPT−4.1	62.0% (4)	68.8% (1)	63.6% (4)	70.0% (4)	48.1% (6)	65.0% (3)	66.7% (3)
Claude 3.7 Sonnet-thinking	62.0% (4)	50.0% (4)	72.7% (2)	65.0% (5)	55.6% (4)	60.0% (5)	100.0% (1)
Claude 3.7 Sonnet	61.0% (6)	50.0% (4)	63.6% (4)	75.0% (3)	55.6% (4)	55.0% (6)	83.3% (2)

Notes: Values are accuracies with performance ranking in parentheses (lower numbers indicate better performance). Human mean = 72.9%, SD = 9.26%. MSK, musculoskeletal.

was \$1.25 / \$10.00. The o3 model, one of the top performers, had a higher price point at \$10.00 / \$40.00. Claude 3.7 Sonnet and Claude 3.7 Sonnet-thinking were both listed at \$3.00 / \$15.00. GPT-4.1 was priced at \$2.00 / \$8.00. These figures, updated as of May 20, 2025, indicate a range of cost structures for accessing these advanced models.

A sample board-style examination used in this study is shown in Figure 6.

DISCUSSION

This study evaluated the performance of six contemporary MLLMs on the 2024 JRS Board Examination, comparing their capabilities against human examinees and assessing the impact of image inclusion. The findings indicate that the most advanced MLLMs, notably Gemini 2.5 Pro Preview and o3, have now surpassed the average performance of fifth–

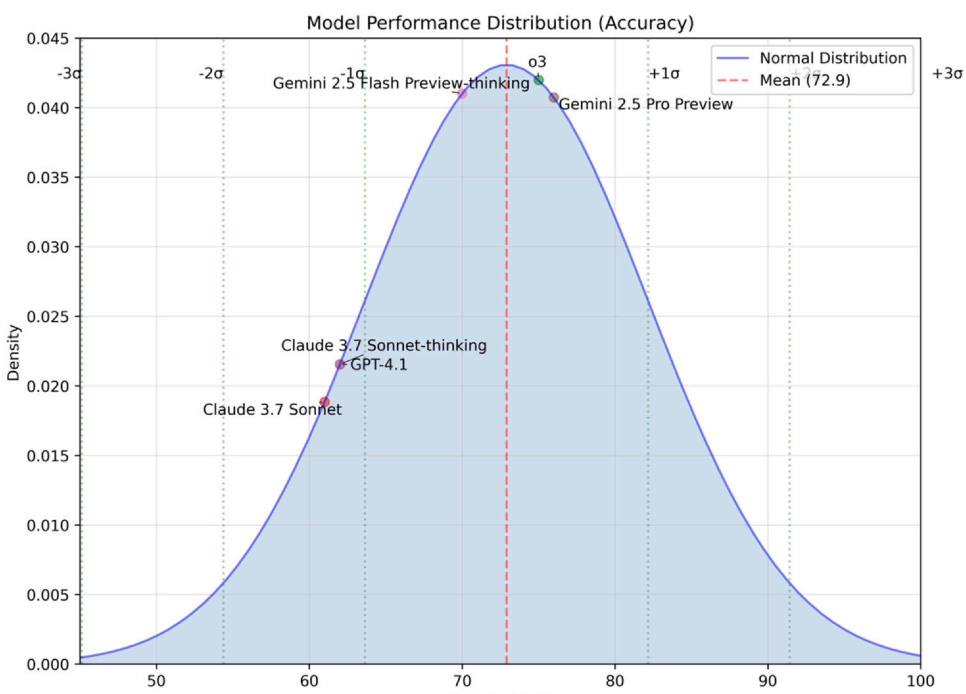

Figure 2. Performance distribution of six multimodal LLMs compared against the normal distribution of human examinee scores on the 2024 Japanese Radiology Board Examination (2nd stage). The blue shaded curve illustrates the normal distribution derived from the accuracy scores of human examinees (fifth-year radiology residents), with a mean accuracy of 72.9% (indicated by the vertical dashed red line) and a SD of 9.26%. Vertical dotted green lines mark integer standard deviations from the human mean (0σ at the mean). The overall accuracy scores for each evaluated LLM are superimposed on this distribution: Gemini 2.5 Pro Preview (76.0%), o3 (75.0%), Gemini 2.5 Flash Preview-thinking (70.0%), Claude 3.7 Sonnet-thinking (62.0%), GPT-4.1 (62.0%), and Claude 3.7 Sonnet (61.0%). This visualization highlights the performance of top-scoring LLMs, Gemini 2.5 Pro Preview and o3, relative to the human average. LLM, large language model; SD, standard deviation.

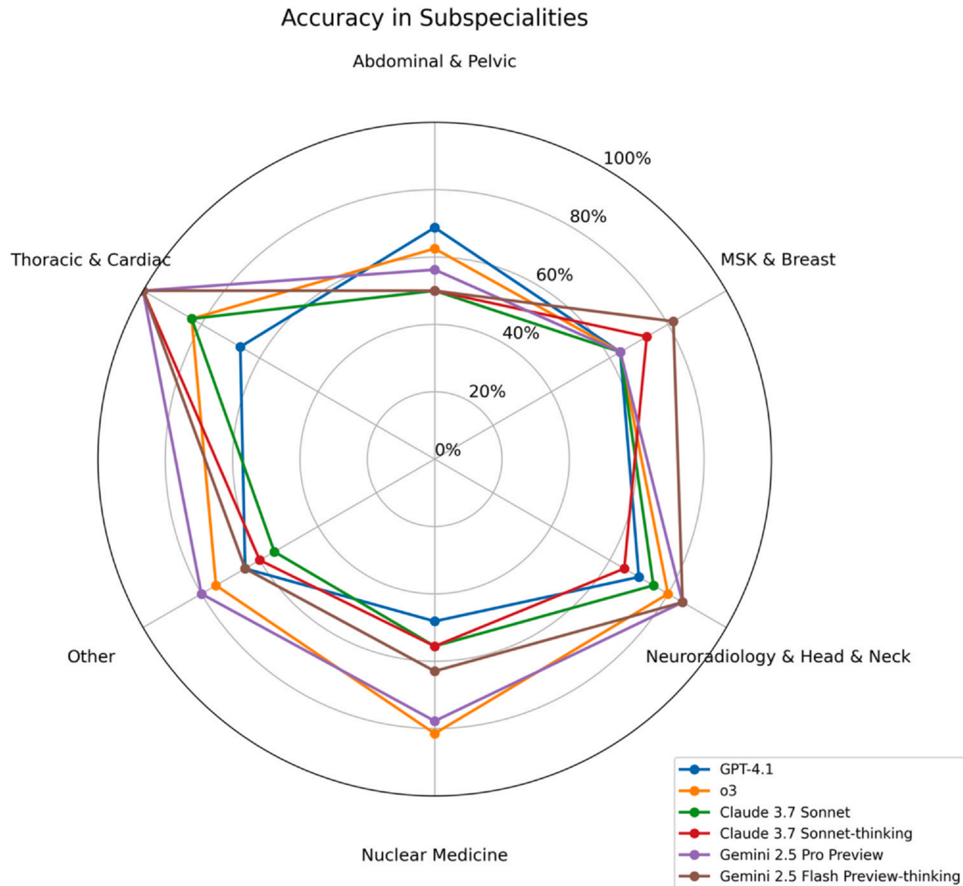

Figure 3. Radar chart comparing the accuracy of six multimodal LLMs across six radiological subspecialties from the 2024 Japanese Radiology Board Examination (2nd stage). Each axis corresponds to a subspecialty: Abdominal and Pelvic Radiology, Musculoskeletal and Breast Imaging (MSK & Breast), Neuroradiology and Head and Neck Imaging, Nuclear Medicine, Other, and Thoracic and Cardiac Radiology. Accuracy is represented by the distance from the center (0%) to the point on the axis (outermost circle representing 100%), with concentric circles marking 20% increments. The performance profile for each LLM is depicted by a unique colored line and marker, as detailed in the chart's inset legend. LLM, large language model; MSK, musculoskeletal.

year radiology residents on this challenging specialty examination. A particularly significant observation was the performance of latest-generation non-reasoning MLLM (Gemini 2.5 Pro Preview) not only achieved the highest overall accuracy but also demonstrated a statistically significant improvement in performance when radiological images were provided at a considerably favorable cost.

The development of MLLMs and the enhancement of their reasoning abilities are central themes in contemporary AI research. Traditional LLMs, while powerful, often operate by identifying statistical patterns and correlations within their vast training datasets. In contrast, models or prompting techniques emphasizing "reasoning" aim to emulate more complex cognitive processes, such as decomposing problems into

TABLE 4. Model Performance Metrics with/without Images

Model Name	Accuracy (with image)	Accuracy (without image)	p-value
GPT−4.1	61.5% (59/96)	63.5% (61/96)	0.824
o3	74.0% (71/96)	70.8% (68/96)	0.607
Claude 3.7 Sonnet	60.4% (58/96)	41.7% (40/96)	< 0.001
Claude 3.7 Sonnet-thinking	60.4% (58/96)	61.5% (59/96)	1.000
Gemini 2.5 Pro Preview	75.0% (72/96)	63.5% (61/96)	0.035
Gemini 2.5 Flash Preview-thinking	68.8% (66/96)	57.3% (55/96)	0.019

Notes: Values are accuracies, with the number of correct answers and total questions in parentheses. A p-value less than 0.05 is considered statistically significant. For Claude 3.7 Sonnet, in 40 out of 100 questions, and for Claude 3.7 Sonnet-thinking, in 11 out of 100 questions, the models responded that they could not provide an answer without images.

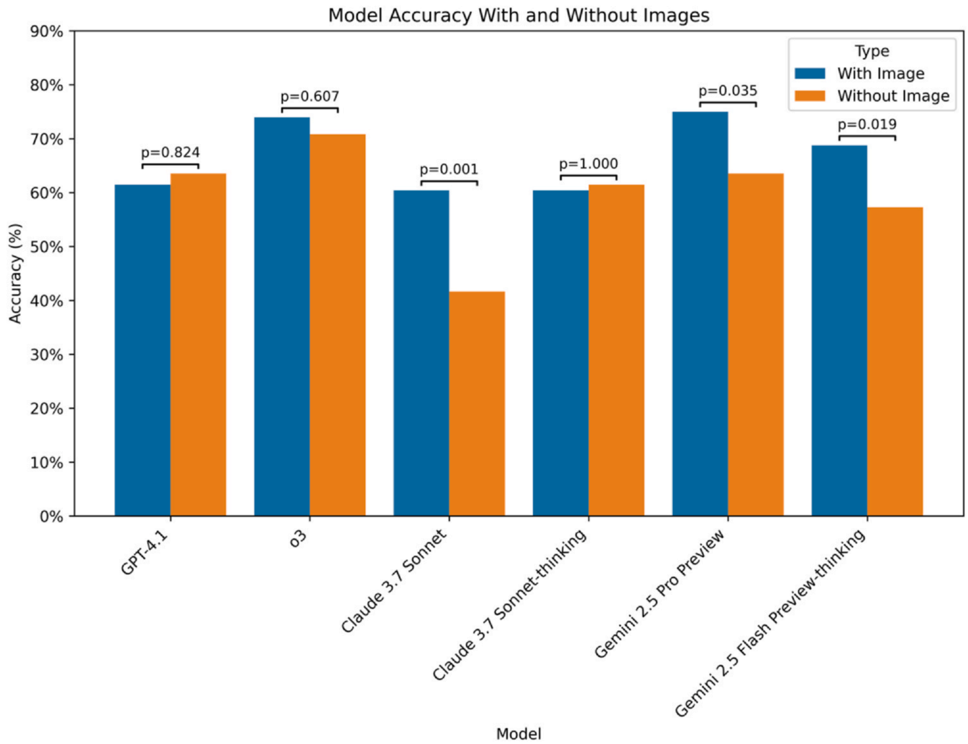

Figure 4. Comparison of accuracy for six multimodal LLMs on the 96 image-based questions from the 2024 Japanese Radiology Board Examination (2nd stage), with and without the inclusion of corresponding images. For each LLM, paired bars display the percentage accuracy achieved when images were provided and when images were omitted, as indicated in the chart's inset legend. The p-values annotated above each pair of bars represent the statistical significance of the difference in accuracy between the "With Image" and "Without Image" conditions for that model, determined using McNemar's exact test; a p-value < 0.05 was considered statistically significant. LLM, large language model.

intermediate steps (e.g., chain-of-thought) (12), performing logical deductions, and maintaining coherence over extended interactions. The merit of such approaches lies in their potential to yield more accurate and reliable outputs, particularly for complex tasks requiring nuanced understanding, and to offer greater transparency if the reasoning steps are

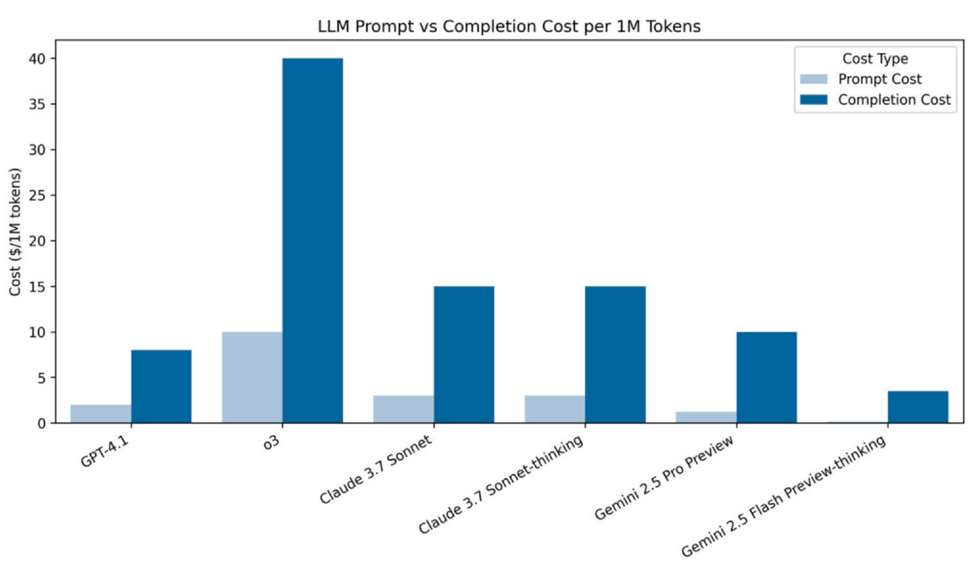

Figure 5. Prompt- and completion-side costs per one million tokens for six leading frontier LLMs on OpenRouter. Bars compare pricing across OpenAI GPT-4.1 and o3, Anthropic Claude 3.7 Sonnet (standard and thinking variants), and Google Gemini 2.5 Pro/Flash Preview models. Prompt costs and completion costs highlight broad price dispersion, with o3 the most expensive ($10/$40) and Gemini 2.5 Flash Preview-thinking the most economical ($0.15/$3.50). LLM, large language model.

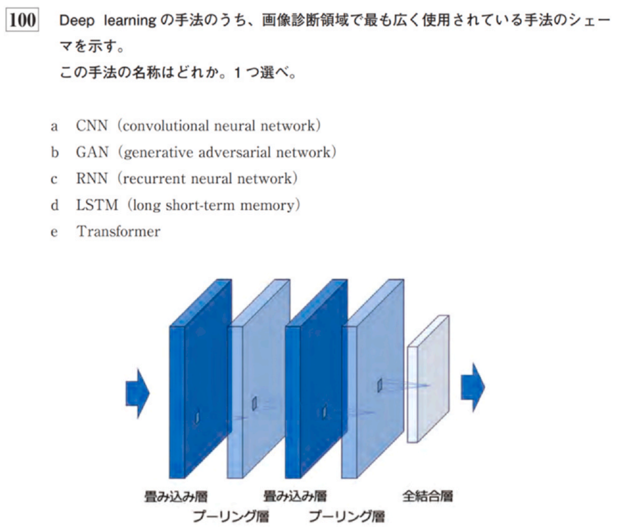

Figure 6. Sample radiology board-style examination. English translation of the stem: “Among deep learning methods, the following schema depicts the technique most widely used in the field of medical image diagnosis. What is the name of this method? Choose one.” The diagram shows stacked convolution and pooling layers followed by a fully connected layer, i.e., a CNN. This item was administered under both image-present and image-absent conditions. Because the stem provided sufficient cues, performance did not depend on image availability: all evaluated MLLMs selected the correct option **(a)** CNN in both conditions. CNN, convolutional neural network; MLLM, Multimodal Large Language Model.

explicit. However, these advanced reasoning processes typically demand greater computational resources during inference, as the model generates and processes these intermediate thoughts or employs more complex attentional mechanisms. This increased computational load often translates to higher operational costs, particularly for large-scale or frequent use (5,13). Encouragingly, the AI field is witnessing a rapid cross-pollination of ideas; insights gained from the development of specialized reasoning techniques are increasingly being integrated into the training and architecture of a broader range of LLMs. This includes advancements in Reinforcement Learning from Human Feedback (RLHF) (14), sophisticated instruction tuning, and the development of MLLMs that learn to process and fuse image and text data more holistically from their inception, rather than treating them as disparate information streams (9,10).

In this study, the outstanding performance of Gemini 2.5 Pro Preview, a model not necessarily requiring explicit multi-step reasoning processes for its baseline operation in this multiple-choice context, is particularly noteworthy. Several factors may contribute to this success. First, its very recent release date (April 2025) suggests it incorporates state-of-the-art architectural designs and training methodologies, likely benefiting from the aforementioned feedback loops from reasoning research and advancements in multimodal training (9,10,12). Its significant performance gain from image inclusion (from 63.5% to 75.0% accuracy) points to a genuinely robust and well-integrated vision-language understanding capability. It is plausible that the inherent capacity of such advanced foundation models has reached a threshold where they can effectively address the complexities of specialized exams like the radiology boards through powerful pattern recognition combined with strong intrinsic reasoning, even without explicit, lengthy chain-of-thought prompting for every question. The clear, structured nature of multiple-choice questions, albeit diagnostically challenging, may also be particularly amenable to the capabilities of such a powerful and well-tuned multimodal system. However, it is worth noting that not all top-performing models demonstrated similar benefits from image inclusion. For instance, GPT-4.1, o3, and Claude 3.7 Sonnet-thinking showed minimal or no accuracy gain. The underlying reasons for this discrepancy remain unclear from our present analysis, and we consider it an important avenue for future investigation.

A critical aspect highlighted by our findings is the cost-effectiveness demonstrated by models like Gemini 2.5 Pro Preview and Gemini 2.5 Flash Preview-thinking. As indicated in Table 2, the pricing for Gemini 2.5 Pro Preview ($1.25/$10.00 per 1 M input/output tokens) is substantially lower than that of o3 ($10.00/$40.00), despite achieving comparable or superior performance. Models or prompting strategies that heavily rely on explicit, verbose reasoning (such as some "thinking" variants or complex chained queries) can incur costs beyond the nominal input and final output token counts. The generation of intermediate reasoning steps, if these are processed or output by the model, also consumes tokens, potentially escalating the true cost per query significantly. For fields like radiology, where AI tools might be used frequently by many professionals for diverse tasks ranging from diagnostic assistance to educational support, a favorable cost-performance ratio is paramount for widespread clinical adoption and integration (15,16). The emergence of models that deliver top-tier performance without incurring premium operational costs represents a significant step towards democratizing access to advanced AI in medicine. These findings suggest potential applications of MLLMs in clinical decision support and training as educational tools offering real-time feedback for residents. However, real-world deployment requires careful validation for safety, reliability, and integration into radiology workflows.

In this study, we found that MLLMs are finally approaching human specialist–level performance at solving image-containing questions; however, this does not imply that radiologists can be immediately replaced in clinical practice. As highlighted in prior work on AI utilization in low-dose CT lung cancer screening (17), real-world radiology involves complex clinical context—history, indications, pretest probability, prior imaging, comorbidities, and downstream management—that cannot be reduced to pixels or algorithmic predictions alone. Accordingly, safe and effective use requires multidisciplinary collaboration among clinicians, data scientists, and ethicists to align model behavior with clinical reasoning and patient values; stepwise, multi-institutional external validation in diverse, real-world cases; continuous post-deployment monitoring with bidirectional human–algorithm feedback to detect data drift, performance degradation, and bias; and explicit human-in-the-loop oversight so that radiologists remain responsible for individualized interpretation and recommendations. Future work should define rigorous frameworks for evaluation, governance, and deployment, and—given the rapid advances in MLLM capability—specify responsible and effective uses across radiology education, research, and clinical decision support.

This study has several limitations that warrant consideration. First, our evaluation was based on a single examination, the 2024 JRS Board Examination (2nd stage). While this is an official and challenging assessment, the findings may not be directly generalizable to other radiological examinations, different question formats (e.g., free-response), or real-world clinical diagnostic tasks, which often involve more ambiguity and a broader range of inputs (15). Second, all questions were originally in Japanese and were instructed to be translated into English by the LLMs (as per their instructions in the prompt). Although the LLMs were tasked with this translation, subtle nuances or cultural contexts might have been lost or altered, potentially affecting their comprehension and performance. Third, the rapid pace of LLM development means that the performance landscape is constantly shifting. Gemini 2.5 Pro Preview, being the newest model evaluated with the most recent knowledge cut-off (January 2025), benefited from the latest advancements. Although exam questions were obtained from a members-only website, reducing the likelihood of public training data exposure, the possibility of inadvertent contamination cannot be fully excluded (16). Consequently, the results presented reflect a snapshot of capabilities as of Spring 2025, and future models or updates to existing ones will undoubtedly yield different performance profiles. Additionally, the "black box" nature of many LLMs makes it challenging to fully understand the underlying reasons for their successes or failures on specific questions beyond analyzing explicit reasoning chains when provided. Fourth, performance on image-omitted questions may partly reflect test-taking heuristics or elimination strategies based on textual cues rather than true image-based diagnostic reasoning. This highlights the need for careful interpretation of text-only accuracy. Fifth, while our results are specific to the JRS examination, similar challenges in multimodal reasoning and image-based decision-making exist in other board exams worldwide. Future validation across ABR, FRCR, and clinical settings is needed. Sixth, this study was unable to present example questions with original patient images, as the examination materials were derived from nationwide, multi-institutional sources, and obtaining informed consent or an opt-out procedure was not feasible. As a result, representative examples of questions and responses could not be disclosed. Seventh, because all prompts involved translation from Japanese to English, subtle linguistic or cultural biases could have influenced model comprehension and performance. Finally, this study primarily focused on diagnostic accuracy; other important aspects, such as the models' confidence calibration, the reliability of their decision-making processes, or the clinical utility of their explanations (if solicited beyond the multiple-choice answer), were not assessed.

In conclusion, this study demonstrates that the latest generation of MLLMs, particularly Gemini 2.5 Pro Preview and o3, can achieve performance levels that exceed the average scores of human specialists on a challenging national radiology board examination. Crucially, models like Gemini 2.5 Pro Preview not only demonstrate a significant ability to leverage image information effectively—a previous bottleneck for MLLMs in radiology—but also offer this top-tier performance with substantial cost advantages.

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CREDIT AUTHORSHIP CONTRIBUTION STATEMENT

Takeshi Nakaura: Conceptualization. **Naoki Kobayashi:** Conceptualization. **Takanori Masuda:** Conceptualization. **Yasunori Nagayama:** Conceptualization. **Hiroyuki Uetani:** Conceptualization. **Masafumi Kidoh:** Conceptualization. **Seitaro Oda:** Conceptualization. **Yoshinori Funama:** Conceptualization. **Toshinori Hirai:** Conceptualization.

DECLARATION OF COMPETING INTEREST

T. Hirai has received research support from Canon Medical Systems. All other authors declare no relevant conflicts of interest.

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DATA-SHARING STATEMENT

Datasets generated or analyzed during the current study are available from the corresponding author on reasonable request.

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